

$$\mathbf{X} = \begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = A \cdot \mathbf{p}(\theta) + \mathbf{W} = \begin{bmatrix} A \cos(\theta) \\ A \cos(2\pi F_0 + \theta) \\ \vdots \\ A \cos(2\pi F_0(N-1) + \theta) \end{bmatrix} + \begin{bmatrix} W[0] \\ W[1] \\ \vdots \\ W[N-1] \end{bmatrix}$$

$$W[n] \in \mathcal{N}(0, \sigma_w^2) \quad IID \quad \rightarrow \quad X[n] \in \mathcal{N}(A \cdot p_n(\theta), \sigma_w^2)$$

$$p_n(\theta) = \cos(2\pi F_0 n + \theta)$$

$$SNR: \quad \gamma = \frac{\|A \cdot \mathbf{p}(\theta)\|^2}{E\{\|\mathbf{W}\|^2\}} = \frac{A^2 \|\mathbf{p}(\theta)\|^2}{E\{\|\mathbf{W}\|^2\}} = \frac{A^2 \frac{N}{2}}{N \sigma_w^2} = \frac{A^2}{2 \sigma_w^2}$$

■ Joint pdf of the observed data:

$$\begin{aligned} f_{\mathbf{x}}(\mathbf{x}; A, \theta) &= \prod_{i=0}^{N-1} f_{x_i}(x_i; A, \theta) = \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_W^2}} e^{-\frac{(x_n - A\cos(2\pi F_0 n + \theta))^2}{2\sigma_W^2}} \\ &= (2\pi\sigma_W^2)^{-N/2} e^{-\frac{1}{2\sigma_W^2} \sum_{n=0}^{N-1} (x_n - A\cos(2\pi F_0 n + \theta))^2} \end{aligned}$$



■ The log-likelihood function (log-LF):

$$\begin{aligned} \ln L(A, \theta) &= \ln f_{\mathbf{x}}(\mathbf{x}; A, \theta) = -\frac{N}{2} \ln(2\pi\sigma_W^2) - \frac{1}{2\sigma_W^2} \sum_{i=0}^{N-1} (x_i - A\cos(2\pi F_0 n + \theta))^2 \\ &= -\frac{N}{2} \ln(2\pi\sigma_W^2) - \frac{1}{2\sigma_W^2} \sum_{n=0}^{N-1} x_n^2 - \frac{A^2}{2\sigma_W^2} \sum_{n=0}^{N-1} \cos^2(2\pi F_0 n + \theta) + \frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \\ &\cong \kappa - \frac{A^2}{2\sigma_W^2} \cdot \frac{N}{2} + \frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \end{aligned}$$



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In fact, if $N \gg 1$:

$$\begin{aligned} \frac{1}{N} \sum_{n=0}^{N-1} \cos^2(2\pi F_0 n + \theta) &= \frac{1}{N} \sum_{n=0}^{N-1} \left[\frac{1}{2} + \frac{1}{2} \cos(4\pi F_0 n + 2\theta) \right] \\ &= \frac{1}{2} \left[1 + \underbrace{\frac{1}{N} \sum_{n=0}^{N-1} \cos(4\pi F_0 n + 2\theta)}_{\cong 0 \text{ if } N \gg 1} \right] \cong \frac{1}{2} \end{aligned}$$

We can also prove that:

$$\frac{1}{N} \sum_{n=0}^{N-1} \sin^2(2\pi F_0 n + \theta) \cong \frac{1}{2}$$

$$\frac{1}{N} \sum_{n=0}^{N-1} \cos(2\pi F_0 n + \theta) \sin(2\pi F_0 n + \theta) = \frac{1}{2} \cdot \frac{1}{N} \sum_{n=0}^{N-1} \sin(4\pi F_0 n + 2\theta) \cong 0$$

■ The **joint ML estimators** of A and θ are obtained by jointly solving the two **likelihood equations**, which are obtained by deriving the log-LF with respect to both the unknown parameters:

$$\begin{cases} \frac{\partial \ln L(A, \theta)}{\partial A} = 0 \\ \frac{\partial \ln L(A, \theta)}{\partial \theta} = 0 \end{cases}$$

$$\begin{aligned} \frac{\partial \ln L(A, \theta)}{\partial A} &= \frac{\partial}{\partial A} \left[-\frac{A^2}{2\sigma_w^2} \cdot \frac{N}{2} + \frac{A}{\sigma_w^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \right] \\ &= -\frac{AN}{2\sigma_w^2} + \frac{1}{\sigma_w^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) = 0 \\ \rightarrow A &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \end{aligned}$$

$$\begin{aligned}\frac{\partial \ln L(A, \theta)}{\partial \theta} &= \frac{\partial}{\partial \theta} \left[-\frac{A^2}{2\sigma_W^2} \cdot \frac{N}{2} + \frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \right] \\ &= -\frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n + \theta) = 0\end{aligned}$$

$$\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n + \theta) = \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n) \cos(\theta) + \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n) \sin(\theta) = 0$$

$$\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n) \sin(\theta) = -\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n) \cos(\theta)$$

$$\frac{\sin(\theta)}{\cos(\theta)} = -\frac{\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)} \rightarrow \theta = -\arctan \left(\frac{\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)} \right)$$

■ Hence, the ML estimator of θ is given by:

$$\hat{\theta}_{ML} = -\arctan \left(\frac{\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)} \right)$$

■ Now, we should insert the ML estimate of θ in the previous expression of A :

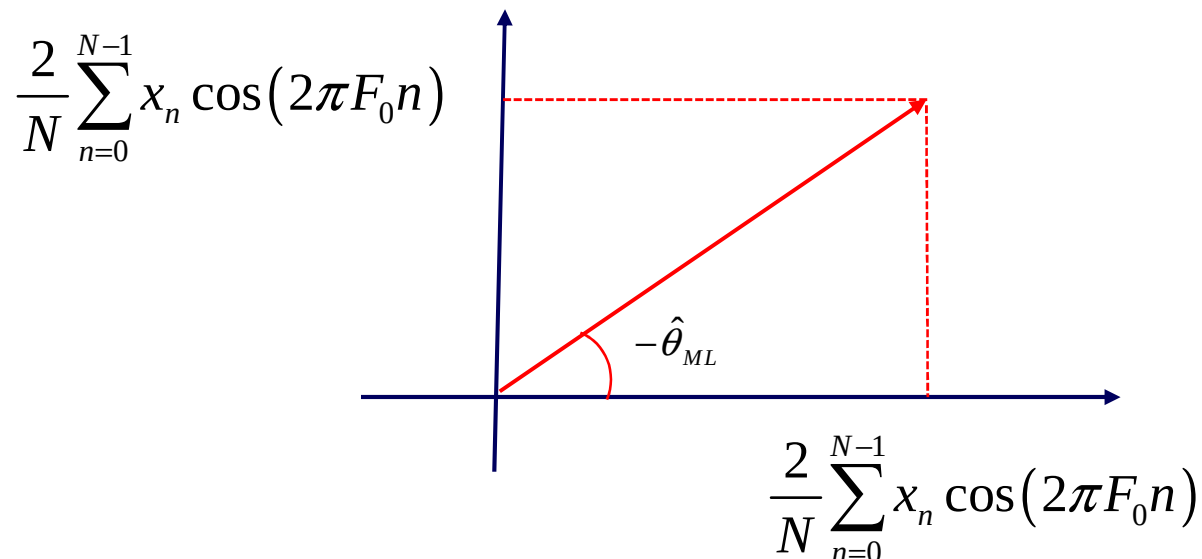
$$\begin{aligned} \hat{A}_{ML} &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \hat{\theta}_{ML}) \\ &= \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n) \cos(\hat{\theta}_{ML}) - \frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n) \sin(\hat{\theta}_{ML}) \end{aligned}$$

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$$\hat{A}_{ML} = \frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n) \cos(\hat{\theta}_{ML}) - \frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n) \sin(\hat{\theta}_{ML})$$

$$\hat{\theta}_{ML} = -\arctan \left(\frac{\sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)} \right) = -\arctan \left(\frac{\frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)} \right)$$



■ Hence, we have:

$$\left\{ \begin{array}{l} \cos(\hat{\theta}_{ML}) = \frac{\frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)}{\sqrt{\left(\frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)\right)^2 + \left(\frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)\right)^2}} \\ \sin(-\hat{\theta}_{ML}) = \frac{\frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)}{\sqrt{\left(\frac{2}{N} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n)\right)^2 + \left(\frac{2}{N} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n)\right)^2}} \end{array} \right.$$

■ The ML estimator of A is immediately derived by inserting these equations in the previous expression of the ML estimator of A .

■ In summary, the **joint ML estimators** of A and θ expressed in terms of the data samples $X[n]$ are given by:

$$\hat{A}_{ML} = \sqrt{\left(\frac{2}{N} \sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n) \right)^2 + \left(\frac{2}{N} \sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n) \right)^2}$$

$$\hat{\theta}_{ML} = -\arctan \left(\frac{\sum_{n=0}^{N-1} X[n] \sin(2\pi F_0 n)}{\sum_{n=0}^{N-1} X[n] \cos(2\pi F_0 n)} \right)$$

■ Note that we are able to implement them only if the frequency F_0 is known.

■ **Cramér-Rao lower Bound (CRB)** on amplitude and phase and comparison of the MSE (derived by Monte Carlo simulation) with the CRB:

$$\begin{aligned} \mathbf{C}_\theta &= E\left\{(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T\right\} = E\left\{\begin{bmatrix} \hat{A} - A \\ \hat{\theta} - \theta \end{bmatrix} \begin{bmatrix} \hat{A} - A \\ \hat{\theta} - \theta \end{bmatrix}^T\right\} \\ &= \begin{bmatrix} E\{(\hat{A} - A)^2\} & E\{(\hat{A} - A)(\hat{\theta} - \theta)\} \\ E\{(\hat{A} - A)(\hat{\theta} - \theta)\} & E\{(\hat{\theta} - \theta)^2\} \end{bmatrix} \end{aligned}$$

$$\mathbf{C}_\theta \geq \text{CRB}(A, \theta) = \mathbf{I}^{-1}(A, \theta) = \begin{bmatrix} -E\left\{\frac{\partial^2 \ln L(A, \theta)}{\partial A^2}\right\} & -E\left\{\frac{\partial^2 \ln L(A, \theta)}{\partial A \partial \theta}\right\} \\ -E\left\{\frac{\partial^2 \ln L(A, \theta)}{\partial \theta \partial A}\right\} & -E\left\{\frac{\partial^2 \ln L(A, \theta)}{\partial \theta^2}\right\} \end{bmatrix}^{-1}$$

■ We have already derived the first order derivatives of the log-likelihood function with respect to the unknown parameters A and θ :

$$\left\{ \begin{array}{l} \frac{\partial \ln L(A, \theta)}{\partial A} = -\frac{AN}{2\sigma_W^2} + \frac{1}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta) \\ \frac{\partial \ln L(A, \theta)}{\partial \theta} = -\frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n + \theta) \end{array} \right.$$

■ The 2nd order derivatives and the **Fisher Information Matrix (FIM)** elements are:

$$\frac{\partial^2 \ln L(A, \theta)}{\partial A^2} = -\frac{N}{2\sigma_W^2} \rightarrow I_{11} = -E \left\{ \frac{\partial^2 \ln L(A, \theta)}{\partial A^2} \right\} = \frac{N}{2\sigma_W^2}$$



$$\frac{\partial^2 \ln L(A, \theta)}{\partial A \partial \theta} = \frac{\partial^2 \ln L(A, \theta)}{\partial \theta \partial A} = -\frac{1}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \sin(2\pi F_0 n + \theta)$$

$$I_{21} = I_{12} = -E \left\{ \frac{\partial^2 \ln L(A, \theta)}{\partial A \partial \theta} \right\} = -\frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} \cos(2\pi F_0 n + \theta) \sin(2\pi F_0 n + \theta) \cong 0$$

$$\frac{\partial^2 \ln L(A, \theta)}{\partial \theta^2} = -\frac{A}{\sigma_W^2} \sum_{n=0}^{N-1} x_n \cos(2\pi F_0 n + \theta)$$

$$I_{22} = -E \left\{ \frac{\partial^2 \ln L(A, \theta)}{\partial \theta^2} \right\} = \frac{A^2}{\sigma_W^2} \sum_{n=0}^{N-1} \cos(2\pi F_0 n + \theta) \cos(2\pi F_0 n + \theta) \cong \frac{A^2 N}{2\sigma_W^2}$$

■ Hence, the FIM is diagonal:

$$\mathbf{I}(A, \theta) = \begin{bmatrix} \frac{N}{2\sigma_w^2} & 0 \\ 0 & \frac{A^2 N}{2\sigma_w^2} \end{bmatrix} \Rightarrow \mathbf{I}^{-1}(A, \theta) = \begin{bmatrix} \frac{2\sigma_w^2}{N} & 0 \\ 0 & \frac{2\sigma_w^2}{NA^2} \end{bmatrix}$$

■ The CRB's on the estimation of amplitude and phase are:

$$\text{var}\{\hat{A}\} \geq \text{CRB}(A) = \frac{2\sigma_w^2}{N} = \frac{A^2}{N\gamma}$$

$$\text{var}\{\hat{\theta}\} \geq \text{CRB}(\theta) = \frac{2\sigma_w^2}{NA^2} = \frac{1}{N\gamma}$$



■ Due to the asymptotic properties of all ML estimators (under some mild conditions), we have that when the data size N goes to infinity, i.e. $N \rightarrow \infty$, or equivalently $\gamma \rightarrow \infty$, we have:

$$\left\{ \begin{array}{l} \hat{A}_{ML} \overset{a}{\in} \mathcal{N}\left(A, \frac{A^2}{N\gamma}\right) \\ \hat{\theta}_{ML} \overset{a}{\in} \mathcal{N}\left(\theta, \frac{1}{N\gamma}\right) \end{array} \right.$$

■ i.e. the estimators are asymptotically efficient, asymptotically unbiased and consistent, since the $\text{MSE} \rightarrow 0$ when $N \rightarrow \infty$ or $\gamma \rightarrow \infty$.